



Y560-A Ammonium ISE Sensor User Manual

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Preface

Dear customer

Thank you for using our product . Reading the entire manual before use is highly recommended for operation and maintenance the instrument and out of unnecessary trouble.

Please observe the operating procedures and precautions in this manual.

To make sure the effective after-sales protection provided by the instrument, please do not use any operation or maintenance other than which mentioned in the manual.

Due to non-compliance with the precautions specified in this manual, any fault and loss caused shall not be covered by the warranty, and the manufacturer shall not bear any relevant responsibility. If you have any questions, please contact our after-sales service department or representative.

Carefully unpack the instrument and accessories from the shipping container, and inspect for possible damage during shipping. Check received parts with items on the packing list. If any parts or materials are damaged or missing, please contact our customer service or the authorized distributor immediately. Save all packing materials until you are sure that the instrument functions properly. Any damaged or defective items must be returned in their original packaging materials.

1 Overview

Yosemitech NH₄-N ISE sensor measures the concentration of dissolved ammonium as nitrogen (NH₄-N) in water. The sensor uses pH, Ammonium Ion electrodes, and an optional potassium ion electrode to determine the NH₄⁺-N concentration. It is designed for use in all kind of natural water, such as lakes, streams, rivers, as well as wastewater plants. The online sensor measures ammonium electron concentration without using any chemicals. With automatic wiper to clean up the electrode sensing surface and easy electrode replacement, Y560-A provides a quick, simple, and economical solution for ammonium concentration measurement.

Features:

RS-485; MODBUS protocol compatible;

No reagent, no pollutants, more economic and environmentally friendly;

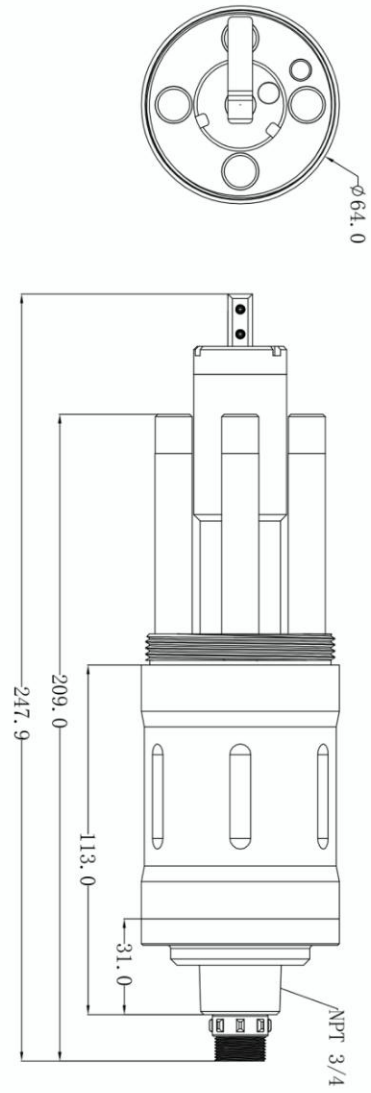
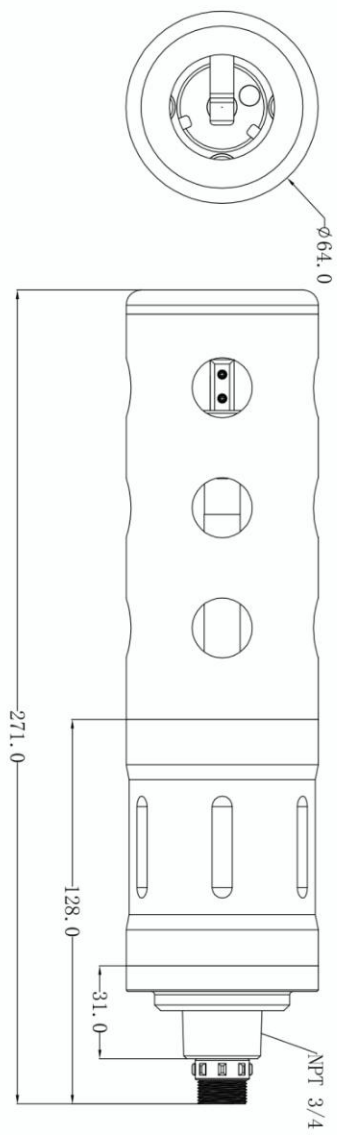
Automatic compensation for NH₄⁺, pH and temperature in water;

With a self-cleaning wiper, prevent biofouling to guarantee accurate measurement.

1.1 Introduction



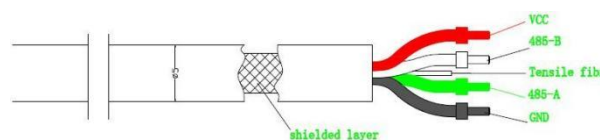
▲ Y560-A Ammonium ISE Sensor



▲ Y560-A Ammonium ISE Sensor Dimensional

1.2 Definition of cable

wire AWG-24 or AWG-26 shielding wire. OD= 5.5 mm



- 1, Red—Power (VCC)
- 2, White—485 Date_B (485_B)
- 3, Green—485 Date_A (485_A)
- 4, Black—Ground (GND)
- 5, Bare wire—shield

1.3 Technical Specification

Name	Ammonium ISE Sensor
Model	Y560-A
NH4_N Range	0-100mg/L (0~1000mg/L optional)
NH4_N Accuracy	±10% or ±0.5mg/L
NH4_N Resolution	0.01 mg/L
pH Range	4-10
pH Accuracy	±0.1pH
pH Resolution	0.01
Size	Φ64mm * 271mm (with protective cover)
Housing IP Rating	IP68
Deepest Depth	10m underwater
Temperature Range	0 ~ 50℃
Sensor Interface	Supports RS-485, MODBUS protocol
Construction	NPT3/4 thread, submersible mounting
Power	DC 9~24V, ≥1A
Cable Length	10m standard, Custom length available
Calibration	one-point or two-points calibration
Sensor housing material	POM

Note:

The above technical parameters are all data under laboratory standard liquid environment.
Sensor life and maintenance calibration frequency are related to actual field conditions.

2 Installation

2.1 Configuration table

Item	Number	Note
NH4-N Sensor	1	
Cable	1	10m
Fitting adapter	1	
NH4CL solution	1	1000mg/L
pH electrode buffer solution	1	20ml
reference electrode buffer solution	1	20ml
ISA ion activator	1	20ml
Rubber protective Cap	1	

2.2 Before use

(1) Take off the protect cap:

Please take off the protect cap of NH4, pH and reference electrode before installation and keep them properly for future use.

(2) Cleaning and activation:

First use DI water to wash the electrodes (DO NOT USE WIPER, IT WILL DAMAGE THE ELECTRODE SENSING FILM).

It needs to be re-activated before use. Activation method: soak the electrode in 10ppm NH4CL solution for more than 24 hours (see standard solution preparation for details).

2.3 Sensor Installation

(1) Wiring and power supply:

- ① The female and male connector of sensor cable should be screwed tightly to avoid moisture incursion
- ② Do not use the sensor cable to pull the sensor! It is required to install sensor in a secure and stable mounting bracket.
- ③ Make sure power supply voltage is correct before power on.

(2) Sensor installation:

- ① It is recommended to install the sensor vertically with electrodes facing down.
- ② Considering water level change, the sensor should be installed 30cm below water level. The sensor should not be installed no more than 2m below water surface for maintenance purpose.
- ③ The sensor must be securely installed to avoid damage caused by water flow and other things.

3 Calibration

3.1 Brief description

Y560-A Ammonium ISE Sensor supports one-point or two-points calibration. The calibration tool can be used with our Smart PC software or according to sensor communication protocol. For our Smart PC software, please scan the QR code on the right to get it. Customers can also develop by themselves according to the communication protocol.



SmartPC User Manual:

- (1) Open the SmartPC and select "Language" in the title bar: English.
- (2) Select the correct port and click "Connect".
- (3) Check calibration to perform calibration operation. Press F1 to get the help document. Find the calibration instructions for the NH₄-N probe.
- (4) You can also use the tool to measure and record data, referring to the help documentation.

3.2 Calibration standard solution and calibration formula

(1) pH standard solution (when pH calibration is required, customized by the customers)

We recommend to purchase the pH standard solution with the value of 4.00, 6.86 and 9.18, respectively.

pH user calibration formula:

$$\text{pH}(\text{standard value}) = K * \text{pH}(\text{measurement}) + B$$

Normal factory default: $K=1$, $B=0$

(2) NH₄⁺ standard solution

Factory supplied with a bottle of NH₄⁺ mother standard solution (NH₄⁺ concentration: 1000mg/L), customer can use the NH₄⁺ mother standard solution to dilute to obtain the corresponding concentration of standard solution;

Notes: before the test, the configured NH₄⁺ standard solution needs to add ISA ion activator and stir well to ensure the ionic strength, and the addition ratio is 2% of the standard solution capacity. The amount of solution required can be immersed in three single-electrode heads.

NH₄⁺ user calibration formula (LG(X) is calculated logarithmic in base 10):

$$\text{LG}[\text{NH}_4^+(\text{standard value})] = K * \text{LG}[\text{NH}_4^+(\text{measurement})] + B$$

Normal factory default: $K=1$, $B=0$

Notes: one-point calibration, keep $K=1$, calculate B , and write K and B at the same time;

two-points calibration, calculate K and B according to the system of univariate quadratic equations, and write K and B at the same time.

(3) Potassium ion standard solution (when equipped with potassium ion electrode)

① Take 1.9067g of KCl and dissolve in deionized water, fixed capacity to 1L. At this time, the 1000mg/L K⁺ mother solution is prepared;

② Take 1mL of the 1000mg/L K⁺ mother solution and dissolve in deionized water, fixed capacity to 1L. At this time, the 1mg/L K⁺ standard solution is prepared;

③ Take 10mL of the 1000mg/L K⁺ mother solution and dissolve in deionized water, fixed capacity to 1L. At this time, the 10mg/L K⁺ standard solution is prepared;

④ Take 100mL of the 1000mg/L K⁺ mother solution and dissolve in deionized water, fixed capacity to 1L. At this time, the 100mg/L K⁺ standard solution is prepared;

Notes: all the above standard solutions need to standardize the use of deionized water to determine the volume, and Wahaha's pure water can be used instead when the preparation conditions of deionized water are insufficient.

K⁺ user calibration formula:

$K + (\text{standard value}) = K * K + (\text{measurement}) + B$

Normal factory default: K=1, B=0

4 Maintenance schedule and methods

4.1 Maintenance schedule

Cleanliness is very important for maintaining accurate readings. The frequency is according to the use environment.

Maintenance task	Recommended maintenance frequency
Sensor cleaning	Cleaning every 2 to 3 weeks
Calibration sensor	Every 3 to 4 weeks
Maintenance and inspection of self-cleaning brush	It is recommended to replace the brush skin every three months (depending on the specific working conditions). return to the factory every 18 months to replace the dynamic sealing ring.

4.2 Maintenance methods

(1) Clean the sensor surface: Wash the outer surface of sensor with tap water, if there is still residue, using soft brush, for some stubborn dirt, household detergent can be added in tap water to clean.

(2) Check the cable: inspect the sensor cable if there is damage.

(3) Electrode Cleaning:

- ① Wash the outer surface of electrode with soft brush. Note that do not touch sensitive membrane of NH_4^+ electrode.
- ② Use clean water to wash the pH and reference electrode. Then gently wipe off with a lint free cloth or a soft brush.
- ③ Do not use anything to wipe sensitive membrane of NH_4^+ electrode. Only rinse it with clean water (DI water is best).
- ④ If sensor needs calibration after cleaning, use a lint free cloth to dry the surface sensor case exclude sensitive membrane. It is recommended to dry by blowing or gently wiping with absorbent paper!
- ⑤ During calibration, electrode cleaning with DI water shall be repeated for each step to avoid polluting the standard solution.

(4) Store the sensor: Regular electrode maintenance requires pH and reference electrode to be stored in protected solutions which equipped with sensor. Please keep NH_4^+ electrode in 1ppm NH_4^+ solution. Note: If the membrane is kept in a dirty or dry state for a long time, it will lead to electrode failure and is not within the warranty scope.

(5) Replace the electrode:

- ① NH_4^+ , reference and pH electrode are all consumable parts. Please replace them in time according to the actual situation.
- ② Change cleaning wiper every 3 months.
- ③ The seal ring of cleaning wiper is guaranteed for one year. It is recommended to send it back to our company for replacement every year.

5 Trouble Shooting

Table 5-1 lists the symptoms, possible causes, and recommended solutions for common problems encountered with the sensor. If your symptom is not listed, or if none of the solutions solves your problem, please contact us.

ERROR	POSSIBLE CAUSE	SOLUTION
Abnormal communication	Controller and cable connection error	Reconnect the controller and cables
	Interface and protocol problems	Use our SmartPC upper computer software to check whether the communication is normal; Check according to the product supporting communication protocol.
No change in value	Abnormal cleaning brush	1.Check whether the brush is entangled by foreign matter, and if so, please remove the foreign matter. 2.Power on again, and observe whether the brush rotates, if it cannot rotate or rotates abnormally. Please contact us.
		Check whether the power supply power meets the requirements.
	Software and hardware anomalies	Please contact us
The measured value is too high or too low.Or the value continues to be unstable.	Sensors are seriously contaminated	Clean the sensor body and special lightwindow surface
	Sensor cleaning brush is worn	Replace the cleaning brush
	Calibration Required	Carry out user calibration
Remaining Issues	Please contact us	

Table 5-1 List of frequently asked questions

6 Warranty Description

(1) Sensor warranty period is 1 year,Electrode warranty period is 3 months .

(2) This quality assurance does not cover the following cases.

- ① Due to force majeure, natural disasters, social unrest, war (declared or undeclared), terrorism, the War or damage caused by any governmental compulsion.
- ② damage caused by misuse, negligence, accident or improper application and installation.
- ③ Freight charges for shipping the goods back to Yosemite Sensing Technology.
- ④ Freight charges for expedited or express shipping of parts or products covered by the warranty.
- ⑤ Travel to perform warranty repairs locally.

(3) This warranty includes the entire contents of the warranty provided by Yosemite Sensing Technology with respect to its products.

① This warranty constitutes a final, complete and exclusive statement of the terms of the warranty, and no person or The agent is authorized to establish other warranties in the name of Yosemite Sensing Technology.

- ② The remedies of repair, replacement, or return of payment as described above are exceptional

cases that do not violate this warranty, and the remedies of replacement or return of payment are for our products themselves. Based on strict liability or other legal theory, Yosemite Sensing Technology shall not be liable for any other damage caused by a defective product or by negligent operation, including any subsequent damage that is causally related to these conditions.

7 Communication protocols

The RS485 communication protocol uses MODBUS communication protocol, and the sensors are used as slaves.

Data byte format.

Baud rate	9600
Starting position	1
Data bits	8
Stop bit	1
Check digit	N

Read and write data (standard MODBUS protocol)

The default address is 0x01, the address can be modified by register

7.1 Reading data

Host call (hexadecimal)

01 03 00 00 00 01 84 0A

Code	Function Definition	Remarks
01	Device Address	
03	Function Code	
00 00	Start Address	See register table for details
00 01	Number of registers	Length of registers (2 bytes for 1 register)
84 0A	CRC checksum, front low and back high	

Slave answer (hexadecimal)

01 03 02 00 xx xx xx xx

Code	Function Definition	Remarks
01	Device Address	
03	Function Code	
02	Number of bytes read	
xx xx	Data (front low and back high DCBA)	See register table for details
xx xx	CRC checksum, front low and back high	

7.2 Writing data

Host call (hexadecimal)

01 10 1B 00 00 01 02 01 00 0C C1

Code	Function Definition	Remarks
01	Device Address	

10	Function Code	
1B 00	Register Address	See register table for details
00 01	Number of registers	Number of read registers
02	Number of bytes	Number of read registers x2
01 00	Data (front low and back high DCBA)	
0C C1	CRC checksum, front low and back high	

Slave answer (hexadecimal)

01 10 1B 00 00 01 07 2D

Code	Function Definition	Remarks
01	Device Address	
10	Function Code	
1B 00	Register Address	See register table for details
00 01	Returns the number of registers written	
7D 2D	CRC checksum (front low and back high)	

7.3 Calculating CRC Checksum

- (1) Preset one 16-bit register as hexadecimal FFFF (i.e., all 1s) and call this register the CRC register.
- (2) Iso-oring the first 8-bit binary data (both the first byte of the communication information frame) with the lower 8 bits of the 16-bit CRC register and placing the result in the CRC register, leaving the upper 8 bits of data unchanged.
- (3) Shift the contents of the CRC register one bit to the right (toward the low side) to fill the highest bit with a 0, and check the shifted-out bit after the right shift.
- (4) If the shifted out bit is 0: repeat step 3 (shift right one bit again); if the shifted out bit is 1, CRC register and polynomial A001 (1010 0000 0000 0001) for the iso-or.
- (5) Repeat steps 3 and 4 until the right shift is made 8 times so that the entire 8-bit data is processed in its entirety.
- (6) Repeat steps 2 through 5 for the next byte of the communication information frame.
- (7) Exchange the high and low bytes of the 16-bit CRC register obtained after all bytes of this communication information frame have been calculated according to the above steps.
- (8) The final CRC register content is obtained as follows: CRC code.

7.4 Register Table

Start address	Command Description	Number of	Data format (hexadecimal)
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		registers	
0x0700H	Get Software and Hardware Rev	2	4 bytes in total 00 ~ 01: hardware version 02 ~ 03: software version For example, reading 0101 represents 1.1
0x0900H	Get SN	7	14 bytes in total 00: reserved 01 ~ 12: serial number 13: Reserved The 12 bytes of the serial number are translated according to ASCII code, i.e. the factory serial number
0x3000H	Device address (read and write) The factory default is 01	1	2 bytes in total 00~01: Device address The range can be set from 1~254 For example, the data obtained is 02 00 (If the low position is in the front, it means that the address is 2) Take address 15 as an example, then 0F 00 Write the corresponding address (low in front) When the current device address is unknown, you can use FF as a common device address to ask for the current
0x3600H	NH4+ User calibration K/B(read/write)	4	Total 8 bytes 00~03: K 04~07: B To read K for example, read out as 4 bytes of data (low bit in front, DCBA format, need to convert this data to floating point, see below for conversion method) To write k, for example, we need to convert k to 32-bit floating point and write it in (DCBA format)
0x1100	pH User calibration K/B(read/write)	4	Total 8 bytes 00~03: K 04~07: B
0x2900	pH original internal factory parameters(read/write) Notes: If the customer only do pH 2 point calibration, This item is not concerned about. But don't change it arbitrarily	12	Total 24 bytes 00~03: K1 04~07: K2 08~11: K3 12~15: K4 16~19: K5 20~23: K6 The default parameter is K1=6.86,K2=6.72,K3=0.04, K4=6.86,K5=-6.56,K6=-1.04。
0x3500H	K+ User calibration K/B(read/write)	4	Total 8 bytes 00~03: K

			04~07: B
0x3400	LGNH4+ Value acquisition	2	Total 4 bytes 00~03: LGNH4+ This value is used for the user calibration
0x2600H	Potential/pH Value acquisition	4	Total 8 bytes 00~03: Potential Value 04~07: pH The reading Potential value/pH value is 4 bytes of data. (The low position is in the front, DCBA format, and this data needs to be converted to a change floating point number. The conversion method is shown below)
0x2800	NH3_N、K+ and NH4+ Value acquisition	6	Total 12 bytes 00~03: NH3_N value 04~07: K+ value 08~11: NH4+ value
0x2400	Temperature value acquisition	2	Total 4 bytes 00~03: Temperature value
0x3100H	Brush startup (write only)	0	Send a write command with a write length of 0
0x3200H	Brush repeated start time setting (read and write)	1	2 bytes in total 00~01: Time Take the reading value 1E 00 (default) as an example, the actual value is 0x001E, that is, 30 minutes. For example, if you need to write for 60 minutes, convert it to 3C 00 for writing.

7.5 Conversion algorithms for floating point numbers

7.5.1 Converting floating point numbers to hexadecimal numbers

Step 1: Convert the floating point representation of 17.625 to binary floating point

First find the binary representation of the integer part

$$17 = 16 + 1 = 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

So the binary representation of the integer part 17 is 10001B

Then find the binary representation of the fractional part

$$0.625 = 0.5 + 0.125 = 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

So the binary representation of the decimal part 0.625 is 0.101B

So the floating point number in binary form for 17.625 expressed in floating point form is

10001.101B

Step 2: Shift to find the exponent.

Shift 10001.101B to the left until there is only one place left before the decimal point to get

1.0001101B, and

$10001.101B = 1.0001101 B \times 2^4$. So the exponential part is 4, which, when added to 127, becomes 131, whose binary representation is 10000011B

Step 3: Calculate the end number

Removing the 1 before the decimal point of 1.0001101B gives the trailing number 0001101B (because the 1 before the decimal point must be 1, the IEEE specifies that only the one after the decimal point should be recorded). An important note for 23-bit trailing numbers: the first bit (i.e. the hidden bit) is not compiled. The hidden bit is the bit to the left of the separator, which is usually set to 1 and suppressed.

Step 4: Symbol bit definition

A positive number has a sign digit of 0 and a negative number has a sign digit of 1, so 17.625 has a sign digit of 0.

Step 5: Convert to floating point

1 digit sign + 8 digits exponent + 23 digits mantissa

0 1000011 00011010000000000000000B (corresponding to 0x418D0000 in hexadecimal)

7.5.2 Converting hexadecimal numbers to floating point numbers

Step 1: Convert hexadecimal number 0x427B6666 to binary floating point number 0100 0010 0111

1011 0110 0110 0110 0110 0110B into sign, exponent and mantissa bits

0 10000100 111101101100110011001100110b

1 digit sign + 8 digits exponent + 23 digits mantissa

Sign bit S: 0 表示正数

Index bit E: $10000100B = 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$
 $= 128 + 0 + 0 + 0 + 0 + 4 + 0 + 0 = 132$

Last digit M: 11110110110011001100110B = 8087142

Step 2: Calculating floating point numbers

$$\begin{aligned} D &= (-1)^S \times (1.0 + M/2^{23}) \times 2^{E-127} \\ &= (-1)^0 \times (1.0 + 8087142/2^{23}) \times 2^{132-127} \\ &= 1 \times 1.964062452316284 \times 32 \\ &= 62.85 \end{aligned}$$



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